Mend-Amar Badral

MendeBadra | Mend-Amar Badral | mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

Skills

Languages:	English (C1 IELTS 8.0)
Programming:	Python, Bash, Julia, C/C++, R
Tools:	Linux, Git, Docker, SSH, pandas, sklearn, PyTorch, opencv, open3d, nerfstudio, Quarto, LaTeX
Technical interests:	Applied mathematics, data analysis, interactive visualization, image processing, 3D computer graphics

Education

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: **Evaluating land degradation using image processing methods**

Experience

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Programming & Image processing:** Developed a custom **Python command line scripts** for processing a total of 1800 multispectral drone imagery dataset capture collected from 5 field areas and calculating vegetation indices and K-means segmentation maps for the vegetation. Utilized multi-threading, made reduction in processing times up to a factor of 5.
- **Photogrammetry:** Produced photogrammetry outputs using OpenDroneMap, with a focus on achieving high-quality reconstruction results through iterative processing and parameter tuning. Configured and validated GPU acceleration in Linux environments, and gained practical experience with Docker-based installation, deployment, and GPU debugging workflows.

Projects

aiSim Data Integration for 3D Gaussian Splatting

- Developed a sensor fusion pipeline to parse and convert over sensor measurements from aiMotive's aiSim ADAS simulation platform into the nerfstudio 3D scene representation format. Worked on the camera coordinate system transformation from sensor frame to the world frame.

Exploratory Data Analysis – Bike Sharing Dataset

- Performed exploratory data analysis on bike sharing dataset. Published the results personal blog website using Quarto.

Deep Learning Foundations and Concepts Book Examples Reproduction

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

Awards

Stipendium Hungaricum Scholarship Sep, 2025